higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PRINCIPLES OF COMPUTER PROGRAMMING

(Second Paper)

NQF LEVEL 3

(10041023)

26 November 2019 (X-Paper)
09:00–13:00

This question paper consists of 18 pages.

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TIME: 4 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Candidates must be seated in the computer laboratory 30 minutes before the start of the examination to ensure that the allocated computer has the appropriate software and can make printouts.

2. Answer ALL the questions.

3. Read ALL the questions carefully. Candidates are allowed 15 minutes reading time.

4. Number the answers according to the numbering system used in this question paper.

5. This question paper is set with programming terms specific to the BASIC programming language and must be completed using VISUAL STUDIO.NET or VB.NET Express, together with MS Office, or any applicable word-processing software.

6. Answer the questions according to the instructions and specifications given in each question. Marks will be awarded according to the set requirements.

7. Answer only what is required in each question, for example, if the question does NOT ask for data validation, then NO marks will be awarded for data validation.

8. Routines such as *search*, *sort* and *selection* must be hard coded by you. You may NOT use the built-in features of the programming language.

9. ALL data structures must be defined by you, unless the question has a partially completed source code.

10. If a question has a partially completed source code, do NOT delete the source code.

11. Candidates are allowed to use the *Help* function of the integrated development environment (IDE).

12. Save your work regularly on the hard disk/flash disk/network drive that you have been given to prevent loss of work. NO additional time may be allowed for work lost due to a lack of saving.

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13. In the event of problems such as a power failure, or a computer or printing device breaking down, the invigilator will make the necessary arrangements to continue with the examination session. Candidates will NOT be penalised for time lost under such circumstances.

14. NO candidate may print work or make a flash drive available for another candidate or access another candidate's work via the network. Any attempt to access any information from or transfer information to another candidate in whatever form is a contravention of the examination rules and regulations, and will be viewed in an extremely serious light.

15. Print your work immediately after completing a question. Your EXAMINATION NUMBER must appear on ALL PRINTOUTS. NO questions without a printed EXAMINATION NUMBER will be marked. Source code and screenshot PRINTOUTS with handwritten EXAMINATION NUMBERS will NOT be accepted or marked.

16. The source code must be printed directly from the integrated development environment (IDE). Candidates are NOT allowed to copy the source code from the IDE and paste it in MS Word for printing purposes.

17. ALL PRINTOUTS of the source code and screenshots printed during the examination session must be submitted to the invigilator.

18. Steps to print a user interface or any other screenshot:

    18.1 Run/Debug the solution.

    18.2 Hold down the ALT key and press the print screen (*Print Scrn*) key.

    18.3 Open a blank MS Word document and paste the screenshot.

    18.4 Insert a blank THREE-column header.

        (a) Add your EXAMINATION NUMBER against the left margin.

        (b) Add your QUESTION NUMBER against the right margin.

    18.5 Make a PRINTOUT of the page.

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19. The following VB.NET solution files will be saved on the computer in the *PCP L3 P2 Examination Year* folder:

19.1 QUESTION 1

File name	Folder name
MedAidCalc.sln	MedAidCalc

19.2 QUESTION 2

File name	Folder name
RainStats.sln	RainStats

QUESTION 1: FUNCTIONS AND SUBPROCEDURES

The following questions refer to the VB.NET solution called *MedAidCalc,* which has been saved on the computer.

A local broad-based black economic empowerment (B-BBEE) company wants to venture into the medical-aid provider industry. They want an application that will assist potential members to calculate how much medical aid costs. You conducted research for the organisation, which produced the following important definitions:



Main member: A person who intends to join or has already joined the medical aid scheme

Adult dependent: A person who is a dependent on the main member's medical aid scheme, namely a:

- spouse or partner; and/or
- parent or parent-in-law (maximum of FOUR parents/in-laws allowed).

Child dependent: A child under the age of 21 years 

The final total monthly contribution paid by a member consists of several costs:

- Price for the main member
- Price for adult dependents
- Price for child dependents

The medical aid prices for each of the above members are also calculated based on the main member's monthly income, as shown in TABLE 1:



INCOME SCALE	MAIN MEMBER PRICE	ADULT PRICE	CHILD PRICE
R6 300 or less	Does not qualify	Does not qualify	Does not qualify
R6 301 to R8 500	R965,00	R965,00	R361,00
R8 500 to R11 700	R1 344,00	R1 344,00	R3 950,00
R11 700 or more	R1 903,00	R1 903,00	R561,00

TABLE 1: INCOME SCALE

NOTE: If a member has THREE or more children, he/she qualifies for a 5% discount on their monthly contributions.



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Below is the IPO-diagram for the problem:

INPUT	PROCESSING	OUTPUT
Monthly income Number of adult dependents Number of child dependents	Basic monthly cost = (main member price + (number of adult dependents * adult price) + (number of child dependents * child price)) If No of Children >= 3 Discount = basic monthly cost * 5% Else Discount =0 End Final monthly cost = basic monthly cost – discount	Monthly contribution Discount

FIGURE 1 is a screenshot of the user interface with descriptive labels of the controls used.

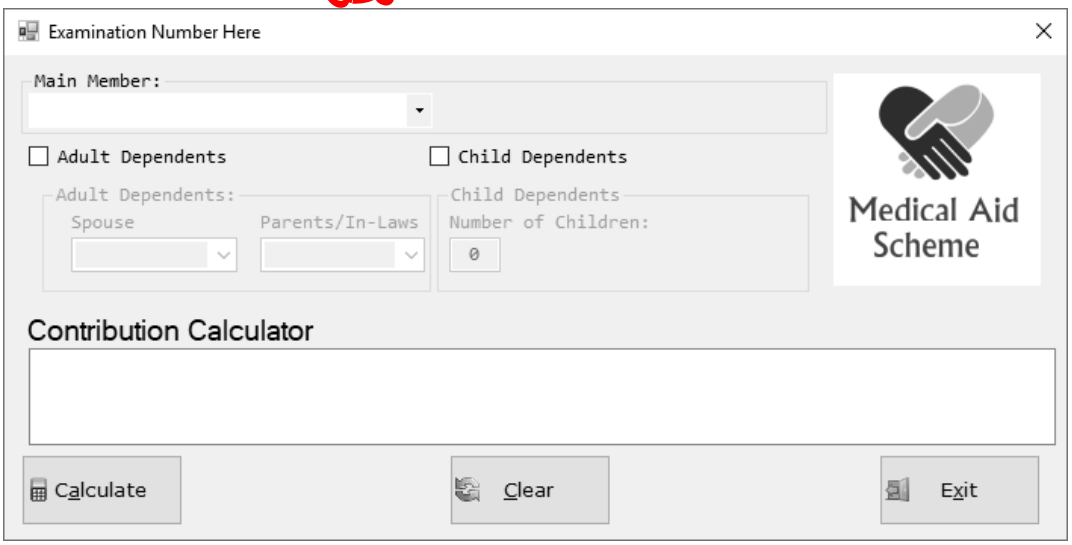
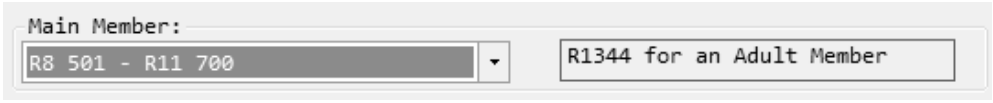


FIGURE 1: MEDAIDCALC – GRAPHICAL USER INTERFACE

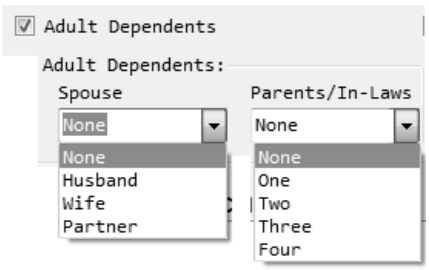
The following steps explain how to use the program to enter data and calculate the total monthly contribution.

STEP 1: Run the application and select an income category. The application will determine the price any adult member has to pay for medical aid.

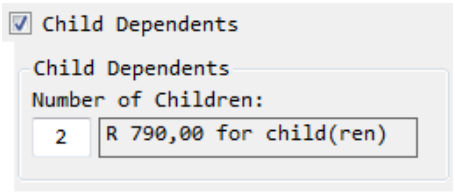


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STEP 2: If the user wants to add any adult dependents, tick the *Adult Dependents* checkbox and select an option presented by the two combo boxes.



STEP 3: If the user wants to add any child dependents, tick the *Child Dependents* checkbox and enter the number of children in the textbox. The application will determine the cost of medical aid for the number of children entered.



STEP 4: When the user has selected the relevant options and clicks on the *Calculate* button, the application will calculate and display the cost of the medical aid.

Main_Member	Adult(s)	Children	Basic_Contribution	Discount	Monthly_Contribution
R 1 344,00	3	3	R 6 561,00	R 328,05	R 6 232,95

The completed user interface with the partially completed source code has been saved on the computer. Open the *MedAidCalc* solution and follow the instructions below to complete the application.

1.1 When the interface *frmCalc* loads:

1.1.1 Invoke the user-defined subprocedure *PopIncomeCategories.* (1)

1.1.2 Declare TWO loop counters and initialise them to ZERO. (2)

1.1.3 Use a pre-test *Do While* loop to populate the *Spouse* combo box with the values of the elements stored in an array called *strArrSpouse*. (5)

1.1.4 Use a *For Next* loop to populate the *Parents* combo box with the values stored in an array called *strArrParents*. (2)

NOTE: The loops for QUESTION 1.1.3 and QUESTION 1.1.4 should use the loop counters declared in QUESTION 1.1.2.

1.1.5 Select the first item in all THREE of the combo boxes. (1)

1.1.6 Make the *Adult Price* label visible. (1)

1.2 Provide the Visual Basic code that will do the following:

When the *Adult Dependents* checkbox is ticked, the *Adult Dependents* group box must be enabled, otherwise the *Adult Dependents* group box must be disabled and the *Spouse* and *Parents/In-laws* combo boxes must be reset. (4)

1.3 Write the Visual Basic code that will, when the *Child Dependents* checkbox is ticked, enable the *Child Dependents* group box and place the *focus* in the *Number of Children* textbox. If the *Child Dependents* checkbox is unchecked, the *Child Dependents* group box must be disabled. (1)

1.4 When the *Main Member* combo box is selected, the application must use a *Select Case* statement to:

1.4.1 Assign an appropriate message to the text property of the *Adult Price* label.

NOTE: For each case, refer to TABLE 1. (1)

1.4.2 Assign appropriate amounts to the variables for the *Main Member* contribution amount, the *Adult Dependents* price and the *Child Dependents* price.

NOTE: For each case, refer to TABLE 1. (3)

1.5 Create a user-defined function called *CalcNoOfAdults*, without parameters to calculate the number of *Adult Dependents*, which will perform the following: (1)

1.5.1 If the main member has selected *Spouse*, the function must assign the number of *Adult Dependents* to 1 otherwise the number should be zero. (2)

1.5.2 If the main member has selected how many *Parents/In-laws* he/she would like to cover, then the function must increment the original number from QUESTION 1.5.1 by the number selected in the *Parents/In-laws* combo box, otherwise the original number from QUESTION 1.5.1 remains unchanged. (2)

1.5.3 Return the *Number of Adult Dependents* to the event procedure that invoked the function.

NOTE: The function procedure must use a procedure-level variable. (1)

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1.6 Create a user-defined function called *CalcMonthlyBasic* to calculate the basic monthly cost of medical aid by doing the following:

1.6.1 The function must accept the following parameters as a copy:

- The cost of the main member's contribution
- The number of adult dependents
- The cost of the adult dependents
- The number of child dependents
- The cost of the child dependents (2)

1.6.2 Return the calculated basic monthly cost to the event procedure that invoked the function. (2)

1.7 Create a user-defined function called *CalcDiscount* to calculate the discount amount for the medical aid.

1.7.1 The function must accept the following parameters as a copy:

- The number of child dependents
- The basic monthly cost (2)

1.7.2 Determine if the main member qualifies for a discount and then return the calculated discount amount to the event procedure that invoked the function. (2)

1.8 When the *Calculate* button is clicked, the application must:

1.8.1 Use a parse method to determine if the number of child dependents entered is a whole number. If NOT, the application must alert the user, highlight and focus the control, and abort the procedure. (3)

1.8.2 Invoke the function procedure to calculate the number of adult dependents and assign the result to an appropriate variable. (1)

1.8.3 Assign the number of child dependents to an appropriate variable. (1)

1.8.4 Invoke the functions to calculate the basic monthly cost and discount amount, and assign the result to appropriate variables. (3)

1.8.5 Calculate the final monthly cost of the medical aid. (1)

1.9 Insert your EXAMINATION NUMBER in the source code and make a PRINTOUT.

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1.10 Change the text in the title bar of the form to your EXAMINATION NUMBER.

Run the application and enter the following test data:

Income: R6 301 – R8 500
Adult dependents: Husband, One parent
Child dependents: None

Click the *Calculate* button and make a screenshot of the output and print. (1)
[45]

QUESTION 2: ARRAYS

The following questions refer to the VB.NET solution called *RainStats,* which has been saved on the computer.

A retired lecturer has kept statistics on monthly rainfall figures since 2014. The rainfall figures in millimetres (mm) are available for the first six months of each year. Now the lecturer would like to:

- Display a table of rainfall figures, as shown in TABLE 2
- Calculate the total rainfall for each year
- Determine the driest year of the five years
- Determine the wettest month of the 30 months (5 years multiplied by 6 months)
- Determine the average rainfall figure per month

	January	February	March	April	May	June
2014	41,2	28,8	15,8	74,8	70,4	70,2
2015	75,6	12,2	35,6	27,2	106,2	67,2
2016	22,2	41,6	22,2	18,2	75,6	52,8
2017	1,6	4,4	15,6	6,4	28,0	5,2
2018	24,2	60,2	105,6	55,4	57,6	81,6

TABLE 2: MONTHLY RAINFALL FIGURES

Below is the algorithm for the problem:

Set the initial values of totals, wettest month, and highest rainfall figure to zero, “” and zero respectively.

- Calculate the total yearly rainfall:

Read the rainfall figure for each month.
Add to the yearly total.
Check for the wettest month.
Repeat the above SIX times for each year.

- Check for the wettest month:

If rainfall figure > highest figure, then set the new highest and wettest month.

- Determine the driest year:

If lowest rainfall > total yearly rainfall, then set the new lowest and year's position.
Repeat the above FIVE times.

- Calculate the average monthly rainfall:

Read the yearly total.
Add to the total.
Repeat the above FIVE times.
Divide the total by the number of months (5 multiplied by 6).

- Display the information:

Total rainfall for each year.
Driest year.
Wettest month.
Average rainfall per month.

The following steps explain how to use the *RainStats* application to calculate the total rainfall figure for each year, check for the wettest month, determine the driest year and calculate the average monthly rainfall figure.

STEP 1: Run the application. The *Rainfall Information* interface will be displayed, as shown in FIGURE 2.

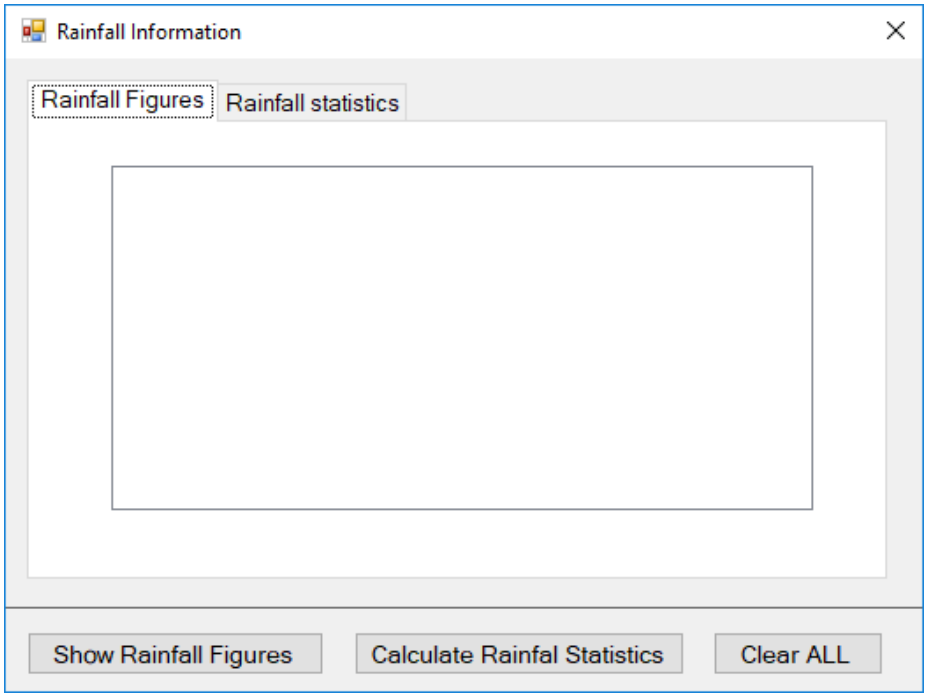


FIGURE 2: RAINFALL STATISTICS GRAPHICAL USER INTERFACE

STEP 2: Click the *Show Rainfall Figures* button to display the table of rainfall figures, as shown in FIGURE 3.

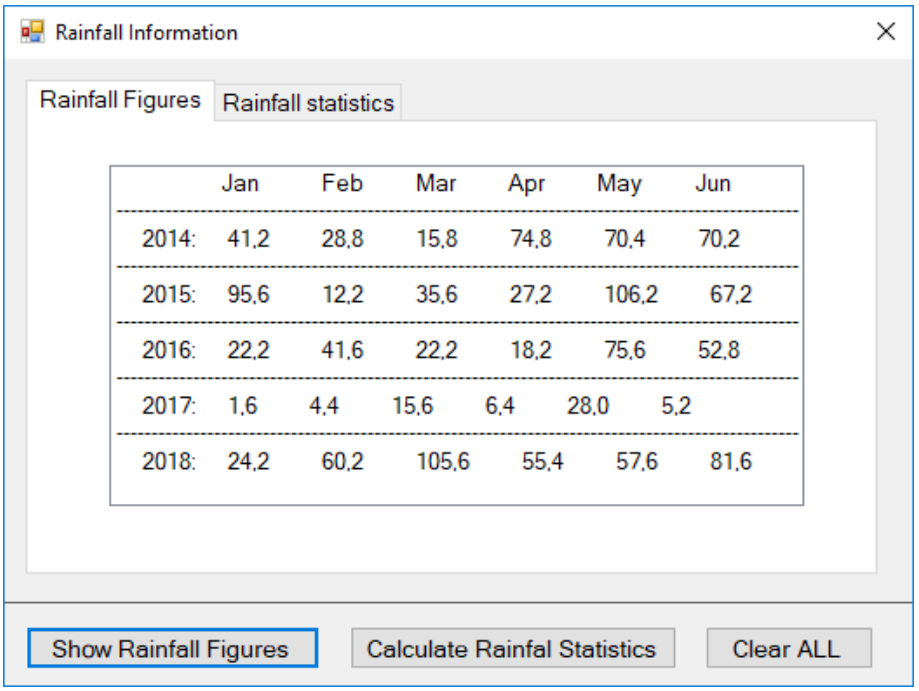


FIGURE 3: TABLE OF RAINFALL FIGURES

STEP 3: Click the *Calculate Rainfal Statistics* button to display the rainfall information, as shown in FIGURE 4.

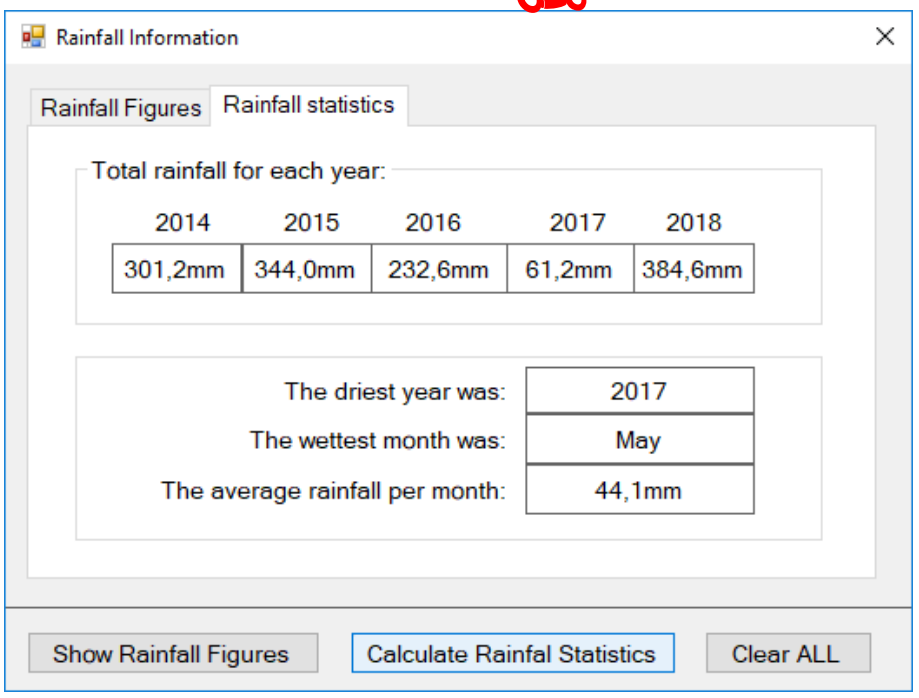


FIGURE 4: RAINFALL INFORMATION

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The partially completed software solution called *RainStats* has been saved on the computer. Open the solution and follow the instructions that follow to complete the application.

2.1 Add the program code to the user-defined subprocedure *Initialisation* to perform the following:

2.1.1 Initialise the one-dimensional array named *sngArrTotalPerYear* to zeroes using a *Do While* loop. (1)

2.1.2 Initialise the *strWettestMonth* and *sngHighestFigure* to a null string and zero respectively. (2)

2.2 Create a user-defined subprocedure called *CheckForWettestMonth* with TWO parameters called *value* and *index* to check for the wettest month.

Add the program code to:

2.2.1 Declare a one-dimensional array to store the following names of months: Jan, Feb, March, April, May and June. (2)

2.2.2 Check if *value* is greater than *sngHighestFigure,* then:

(a) Assign *value* into *sngHighestFigure*

(b) Set *strWettestMonth* to an appropriate month (3)

2.3 Create a user-defined function called *CalcYearlyTotalRainfall* with ONE parameter array called *myArray* to determine the total rainfall for the specific year. (1)

Add the program code to:

2.3.1 Use a *For Next* loop to determine the total monthly rainfall figures and invoke the procedure to check for the wettest month. (4)

2.3.2 Return the total rainfall. (1)

2.4 Add program code to the user-defined subprocedure called *CalcRainfallTotals* to calculate the rainfall totals for each year, by doing the following:

Invoke the *CalcYearlyTotalRainfall* with an array containing rainfall figures as its argument and then assign its value into an array named *sngArrTotalPerYear*.

NOTE: s*ngArrTotalPerYear* is the storage for all rainfall totals. (3)

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2.5 Create a user-defined function called *DetermineDriestYear* without parameters to return the driest year to the calling procedure.

2.5.1 Declare a one-dimensional array to store: '2014', '2015', '2016', '2017' and '2018'. (1)

2.5.2 Implement the sequential search technique to find the lowest rainfall total. (4)

2.5.3 Return the driest year. (1)

2.6 Create a user-defined function called *CalcMonthlyAvgRainfall* without parameters to calculate the average monthly rainfall.

2.6.1 Multiply the number of years by the number of months to determine the overall number of months the rainfall figures were kept. (1)

2.6.2 Determine the sum of all the rainfall totals from the array called *sngTotalRainfallPerYear*. (3)

2.6.3 Use a *Try Catch* block to calculate the monthly average rainfall. (2)

2.6.4 Return the monthly average rainfall.

NOTE: The function procedure must use procedure-level variables. (1)

2.7 Insert your EXAMINATION NUMBER in the source code and print.

[30]

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QUESTION 3: DATABASE

The SRC of your college has established a cold drink stand during registration with the aim of raising funds to buy sanitary towels for disadvantaged students identified by Student Support Services.

A former IT student volunteered to design the TWO database tables named *Product* and *Sales* (refer to TABLE 3 and TABLE 4) and to show the relationship between the tables.

Use the given information to set up the database tables in MS Access. The database design consists of TWO tables. Open a new blank database and save it as *SRC* together with your EXAMINATION NUMBER, for example *SRC-940426XXXXXX1*.

Create the *Product* table, as indicated below, and save the table as *tblProduct*.

Field name	Data type	Field size	Description
prodCode	Text	4	Unique code (primary key)
prodName	Text	20	Product's name
prodPrice	Number	Double	Selling price
prodQtyInStock	Number	Integer	Number of products at hand

TABLE 3: PRODUCT

Create the *Sales* table as indicated below and save the table as *tblSale*.

Field name	Data type	Field size	Lookup	Indexed	Description
saleID	Number	Integer		Yes No Duplicates	Sale ID number
saleTime	Date/Time	Short time			Time of sale
saleItem	Text	20	Product code, Product name		Sold item's name (foreign key)
saleQty	Number	Integer			Number of items bought

TABLE 3: SALES

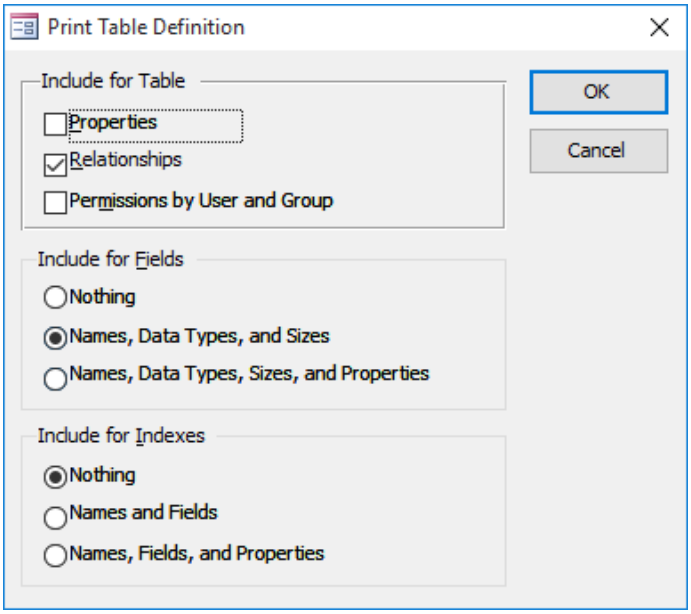
Enforce referential integrity on the relationship already created between the TWO tables.

3.1 Open the *Sales* table in design view and perform the following:

3.1.1 Select the *saleID* field and click on the *General* tab. Make a screenshot of the table's design and print. (1)

3.1.2 Select the *saleItem* field and click on the *Lookup* tab. Make a screenshot of the table's design and print. (1)

3.2 Set the following options for the tables using the database documenter:



Print the table definitions for the TWO tables. (8)

3.3 Follow the instructions below to capture all input data into the database tables.

3.3.1 Open the *Product* table in datasheet view and insert the following records:

prodCode	prodName	prodPrice	prodQtyInStock
P001	Coke 330 mℓ	R11	150
P002	Fanta 330 mℓ	R11	100
P003	Sprite 2 ℓ	R18	200
P004	Coke 2 ℓ	R19	150

3.3.2 Use the form wizard to create a simple form with columnar layout, as shown in FIGURE 5, based on the *Sale* table.

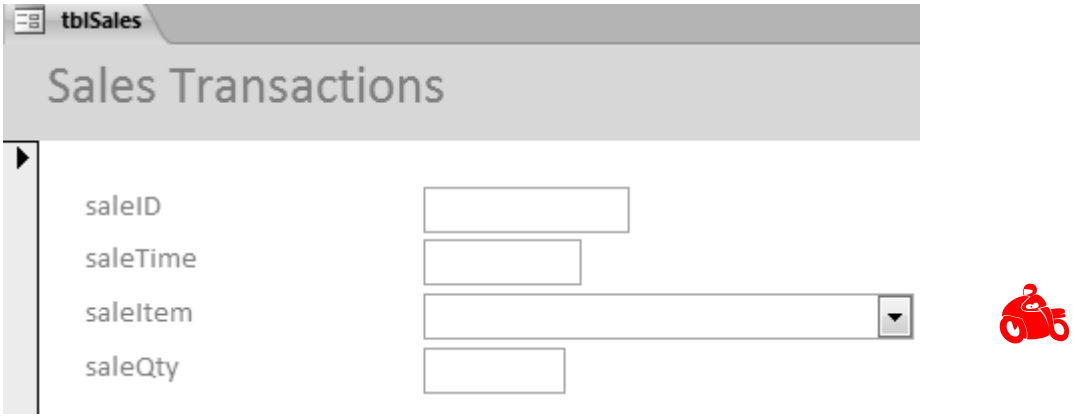


FIGURE 5: SALES-CAPTURING FORM

3.3.3 Make a screenshot of the form and print. (2)

3.3.4 Use the simple form created in QUESTION 3.3.2 to insert the following sales records:

saleID	saleTime	saleItem	saleQty
1010	09:30	Fanta 330 mℓ	1
1011	09:36	Coke 330 mℓ	2
1012	09:45	Coke 2 ℓ	2
1013	10:00	Coke 330 mℓ	3
1014	10:30	Fanta 330 mℓ	1
1015	10:35	Coke 2 ℓ	2
1016	10:48	Coke 330 mℓ	2

3.4 Create a database query based on both tables using ALL the instructions given below.

3.4.1 Select the following fields:

Product table: Product name
Sales table: Sale quantity
Product table: Product price

3.4.2 Add a new field called *TotalCost* and calculate the total cost as follows: Product Price * Sale Quantity.

3.4.3 Change each field name to the new name given in brackets:

- prodName (Item Name)
- saleQty (Quantity)
- prodPrice (Unit Price)

3.4.4 Save the query as *qrySale* and your EXAMINATION NUMBER.

3.4.5 Run the query and print the resulting records. Ensure that ALL columns are clearly visible. (3)

3.4.6 Open the query in design view and make a screenshot of the query design and print. Ensure that ALL columns are clearly visible. (3)

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3.5 Create a database report based on the query created in QUESTION 3.4. Follow ALL the instructions given below.

3.5.1 The report must display ALL fields from the query. (1)

3.5.2 Group by Item Name. (1)

3.5.3 Report layout: Stepped and portrait. (1)

3.5.4 Report title: Sales Data Report and your EXAMINATION NUMBER. (1)

3.5.5 Open the report in design view and adjust the width of ALL columns to appropriate sizes.

3.5.6 Add a formula to calculate the total cost for each sales record. (1)

3.5.7 Add 'SUBTOTAL' as a descriptive label to the formula created in QUESTION 3.5.6. (1)

3.5.8 Add the currency symbol (R) to the total costs. (1)

3.5.9 Make a PRINTOUT of the report.

[25]

TOTAL: 100

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